



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201- Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Module III / Indexing, Selection, Filtering
- **Course Outcome:** CO3
- **Topic Learning Outcome:** -
- **Activity Chosen:** One Minute Paper
- **Justification:**

Selecting **indexing, selection, and filtering** for a One-Minute Paper is justified because these operations are the functional "building blocks" of data manipulation where students frequently encounter **conceptual confusion** between position-based and label-based logic. Since these tasks require more than just memorizing syntax—specifically, the ability to apply **Boolean logic** and understand data structure navigation—the activity provides an immediate "dipstick" check to identify **muddy points** before moving to advanced analytics. It forces students to quickly synthesize how they retrieve specific data subsets, ensuring they have the **foundational accuracy** required for error-free data cleaning and real-world problem solving.

Time Allotted for the Activity: 01 Minutes

Details of the Implementation:

- The time allotted for the topic would remain **01 Minute**.
- After completing the topic explanation and practical demonstration, the following activity was conducted.
- The students would be given the specific prompt: **"Write the Python Script to find the followings. Score=[150,180,50,75,320,200,75] i. find the 5th element ii. Select the 3rd to 5th element iii. Print the score below 100"**
- Once the prompt is given, students quickly write down the rules for **indexing** and **filtering** in their notebooks.
- Most students finish within the minute, sketching out **code snippets** for selecting rows or columns and applying simple filters.
- They focus on writing the **exact syntax** (like `.loc` or `.iloc`) in a short and clear way to show they understand the logic.

- The completed papers provide a quick snapshot of learning, which can be collected from across the classroom to check how well the whole group grasped the topic.

- CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PSO1	PSO2	PSO3
CO3	3	3	3	3	3	3	3

(1 – Low

2 – Moderate

3 – High)

- PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO5	PSO1	PSO2	PSO3
	3	3	3	3	3	3	3
Justification for Correlation	The students will apply Using logical indexing and Boolean algebra to pinpoint specific data points within large structures.	Students will be able to Identifying the correct selection method (position-based vs. label-based) to solve data retrieval challenges.	Students will be able to provide Designing efficient filtering scripts to clean and prepare datasets for further analysis.	Students will be able to Mastering the use of specialized libraries (like Pandas or NumPy) and their specific syntax for data manipulation.	The students will have the ability to perform complex data slicing and sub setting, which is a core skill in data-driven programming	The students able to apply conditional logic (filtering) to translate real-world business requirements into executable code.	The students will develop the speed and accuracy in data handling required for professional roles in data analytics and software development.

- Images / Screenshot of the practice:**

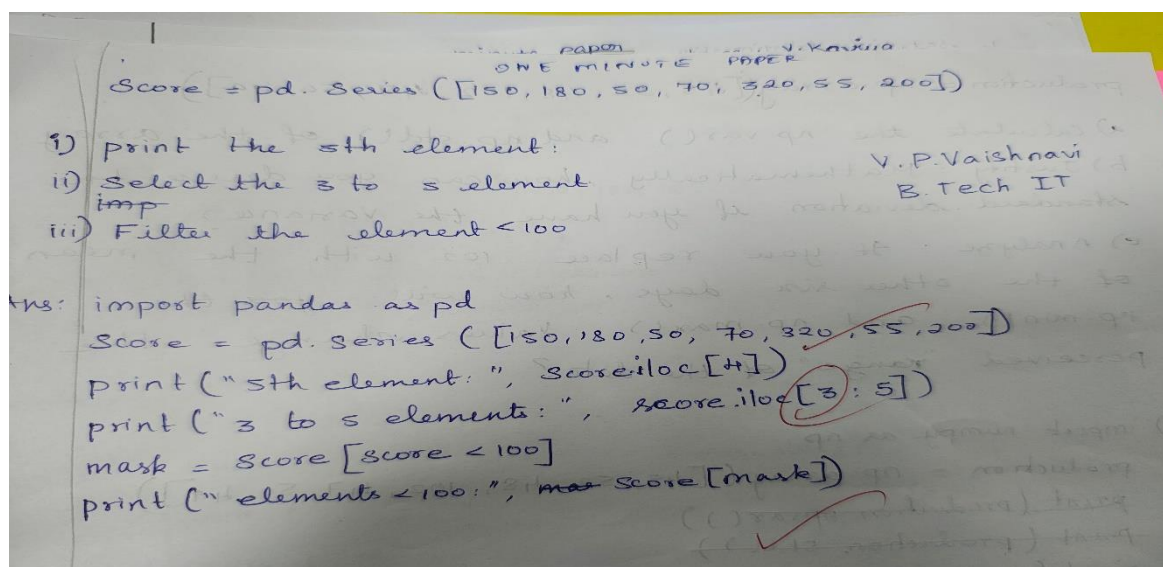


Figure 1. a One minute paper of V.P. Vaishnavi

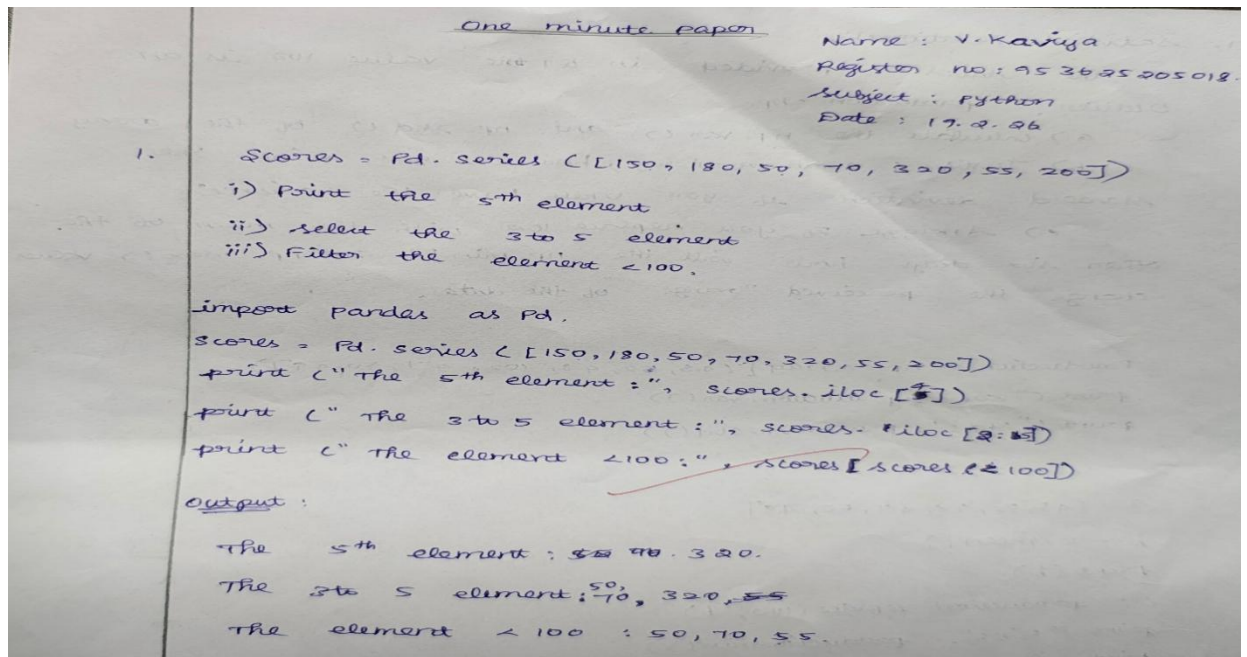


Figure 1. b One minute paper of V.Kaviya

❖ **Reflective Critique:**

- The One-Minute Paper worked as a fast and effective way to check if students could actually write Pandas code on their own.
- The time limit made students focus on the most important parts: indexing, slicing, and filtering logic.
- It helped the teacher quickly see that while students knew how to find a single element, they struggled more with exclusive slicing (like [2:5]).
- Overall, this activity successfully turned a lecture into a practical, timed challenge that reinforced data manipulation skills.

❖ **Benefit of the practice:**

- The quick writing burst kept students focused and made them take the lead in their own learning.
- It helped students see if they truly understood zero-based indexing or were just following the screen.
- Mentally organizing the logic for slicing and filtering helped students remember the Pandas commands better.
- Students quickly identified where they were stuck, allowing the teacher to fix syntax errors immediately.

❖ ***Challenges faced in implementation:***

- It was hard for some to recall the exact rules for slicing and write the code in only 60 seconds.
- Students often mixed up square brackets or missed symbols while trying to finish the code quickly.
- Translating "3rd to 5th" into the correct computer indices (2 to 5) was tricky under pressure.
- Because the clock was ticking, some students finished the indexing but did not have time to complete the filtering part.

References:

https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3452_OneMinutePaper.pdf

https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/MRK_CCS334_One_minute_paper_U2.pdf

Signature of Faculty Member

HOD